

Paired 7/28-day compressive-strength model

30 mixtures × 2 curing ages · linear mixed model (random mixture intercept, age-specific residual variance) · corrected data set · generated 2026-09-29

Findings

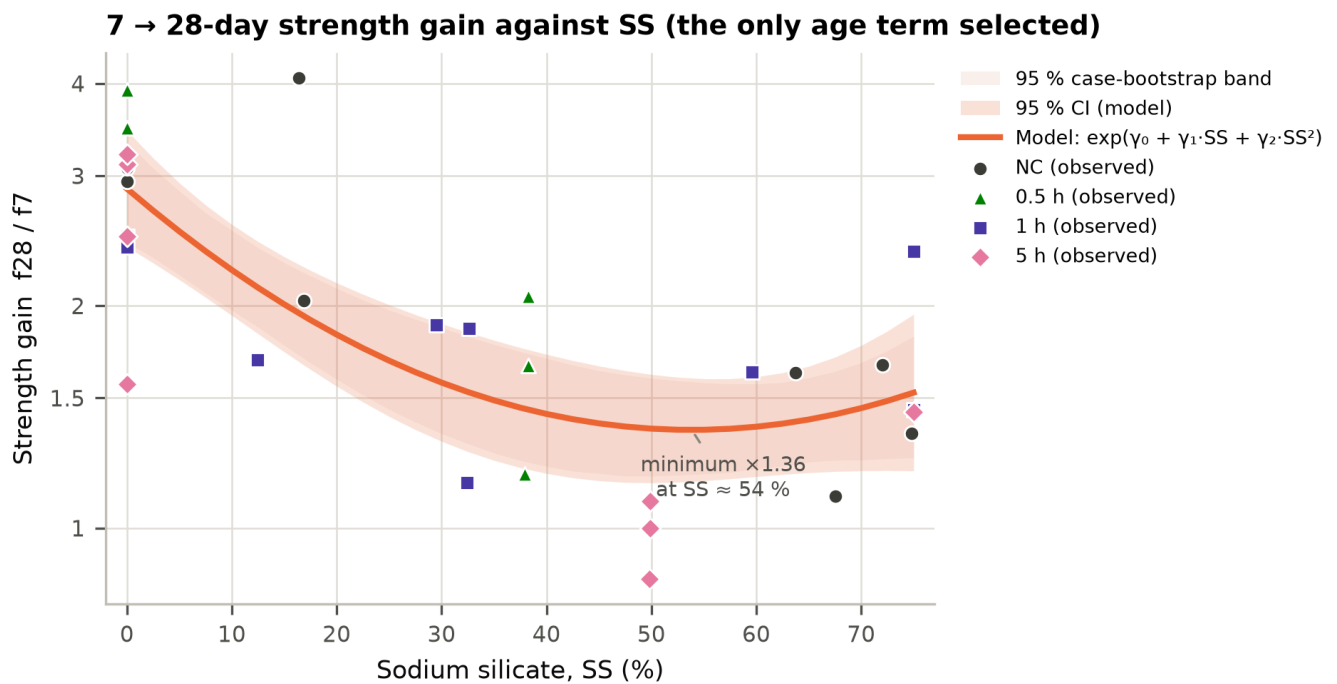
Robust · Strength rises with SS to a peak at mid-range SS, at both ages SS and SS ² are the largest terms (p < 0.001, p < 0.001) and are kept in at least 99 % of re-selections. The peak lies at SS 58 % (7 d) and 64 % (28 d) for uncarbonated mixes at RCF 30 %, and lower (SS 46 % at 7 d, 37 % at 28 d) for carbonated mixes at RCF 50 %.
Robust · The 7 → 28-day gain is set by SS $f_{28}/f_7 = 2.88$ (95 % CI 2.40–3.45) at SS 0 %, falls to a minimum of 1.36 near SS 54 % and is 1.53 (1.20–1.95) at SS 75 % (SS × Age p < 0.001). NaOH-only mixes gain most. SS is the only age-dependent variable in 86 % of residual-bootstrap and 89 % of subsample re-selections. The curvature of the gain (SS ² × Age, p = 0.008) is moderate.
Moderate · Carbonated RCF helps in NaOH-rich mixes and hurts in silicate-rich ones Carbonated × SS p < 0.001. At RCF 50 % and SS 0 % a 1 h carbonated mix is ×1.58 (95 % CI 1.22–2.06; p = 0.002) the uncarbonated one; at RCF 10 % and SS 75 % it is ×0.53 (95 % CI 0.40–0.70; p < 0.001). The crossover SS rises with RCF, from 5 % at RCF 10 % to 50 % at RCF 50 % (1 h). Kept in 98 % of residual-bootstrap but only 71 % of subsample re-selections: it rests on the 8 uncarbonated mixes.
Moderate · RCF raises strength when it is carbonated and when SS is low Carbonated × RCF p = 0.015, RCF × SS p = 0.019. Going from 10 to 50 % RCF multiplies strength by ×2.12 (95 % CI 1.69–2.65; p < 0.001) for carbonated RCF at SS 0 %, and by ×0.90 (95 % CI 0.66–1.24; p = 0.51) for uncarbonated RCF at SS 75 %. RCF never lowers strength significantly. The case-bootstrap intervals of carbonated × RCF and RCF × SS include zero, so these two terms depend on a few mixtures.
Suggestive · Carbonation duration (0.5, 1, 5 h): no difference in level; a smaller 28-day gain after 5 h is possible The three carbonated levels do not differ in level (p = 0.098); on/off coding fits about equally well (ΔAICc –0.83). Carbonation × age is suggestive but not selected (added-term p = 0.081, ΔAICc +4.1). In the age-specific fit the 5 h level gains less than NC (×0.75, 95 % CI 0.57–0.98; p = 0.038), a single contrast that the joint test does not support. The only mixes that do not gain strength from 7 to 28 days are 5 h mixes (mix 17, 5 h, RCF 36.6 %, SS 49.88 %: 16.90 → 16.88 MPa; mix 28, 5 h, RCF 36.6 %, SS 49.83 %: 20.82 → 17.75 MPa).
Not detected · A/B ratio (0.42–0.48): no effect detected No A/B term improves the model at either age (A/B p = 0.74, (A/B) ² p = 0.26, A/B × Age p = 0.29). A/B curvature is selected only if the 28-day variance is forced equal to the 7-day variance (CS); under nested cross-validation that search predicts 7-day strength clearly worse (R ² 0.76 vs 0.89) for a small gain at 28 days (0.14 vs -0.09).

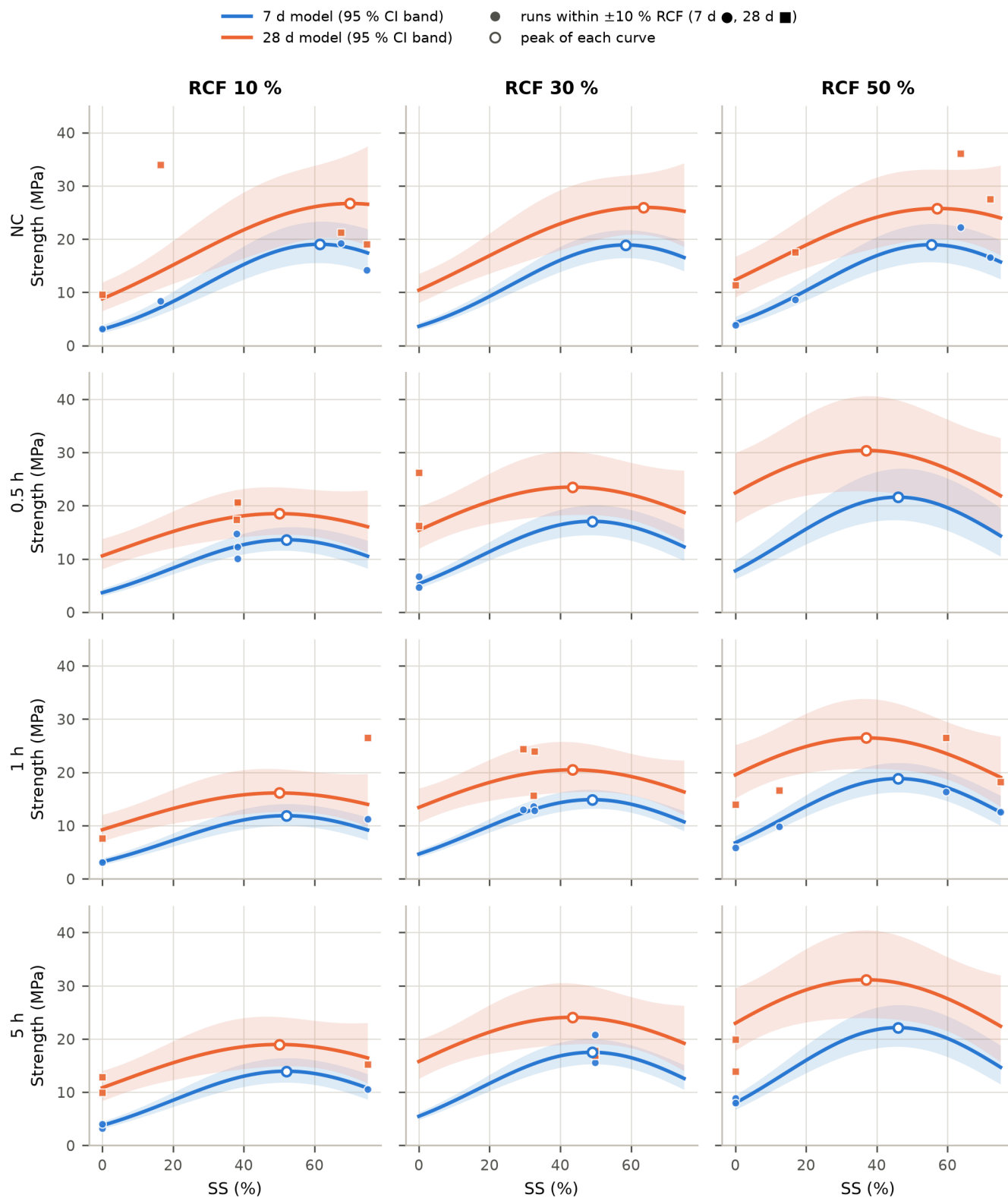
Final model

$\ln f = 2.9151 + 0.0591 \cdot A + 0.6010 \cdot D + 0.0465 \cdot C_{0.5h} - 0.0915 \cdot C_{1h} + 0.0705 \cdot C_{5h} - 0.1102 \cdot AD - 0.4991 \cdot D^2 + 0.2057 \cdot K \cdot A - 0.3441 \cdot K \cdot D + g \cdot (0.3752 - 0.3179 \cdot D + 0.3648 \cdot D^2)$			
$\ln(f_7) = 1.03206 + 0.00846427 \cdot RCF + 0.0610224 \cdot SS - 1.4687 \times 10^{-4} \cdot RCF \cdot SS - 4.8466 \times 10^{-4} \cdot SS^2 + 0.0819631 \cdot C[0.5 \text{ h}] - 0.0559958 \cdot C[1 \text{ h}] + 0.105909 \cdot C[5 \text{ h}] + 0.0102872 \cdot K \cdot RCF - 0.00917525 \cdot K \cdot SS$			
$\ln(f_{28}) = 2.09 + 0.00846427 \cdot RCF + 0.0330868 \cdot SS - 1.4687 \times 10^{-4} \cdot RCF \cdot SS - 2.2523 \times 10^{-4} \cdot SS^2 + 0.0819631 \cdot C[0.5 \text{ h}] - 0.0559958 \cdot C[1 \text{ h}] + 0.105909 \cdot C[5 \text{ h}] + 0.0102872 \cdot K \cdot RCF - 0.00917525 \cdot K \cdot SS$			
$\ln(f_{28}/f_7) = 1.05795 - 0.0279356 \cdot SS + 2.5943 \times 10^{-4} \cdot SS^2$			
7 d · R ² 0.96 pred. R ² 0.91 · RMSE 2.1 MPa	28 d · R ² 0.23 pred. R ² -0.06 · RMSE 7.6 MPa	Residual scatter 15 % / 35 % 7 d / 28 d (replicate 18 % / 19 %)	7 ↔ 28 d correlation 0.52 within mixture (paired)

Term	df	F	p	Residual bootstrap	Subsamples 24/30	Evidence
RCF	1, 20.0	0.89	0.36	100 %	100 %	Retained for hierarchy
SS	1, 26.6	67.26	< 0.001	100 %	100 %	Robust
Carbonation (4 levels)	3, 20.0	1.75	0.19	99 %	75 %	Retained for hierarchy
RCF × SS	1, 20.0	6.57	0.019	84 %	52 %	Moderate
SS ²	1, 20.2	25.38	< 0.001	100 %	100 %	Robust
Carbonated × RCF	1, 20.0	7.02	0.015	85 %	46 %	Moderate
Carbonated × SS	1, 20.0	19.32	< 0.001	98 %	71 %	Moderate
Age (28 vs 7 d)	1, 27.0	16.96	< 0.001	-	-	Design factor
SS × Age	1, 27.0	19.77	< 0.001	100 %	100 %	Robust
SS ² × Age	1, 27.0	8.10	0.008	83 %	79 %	Moderate

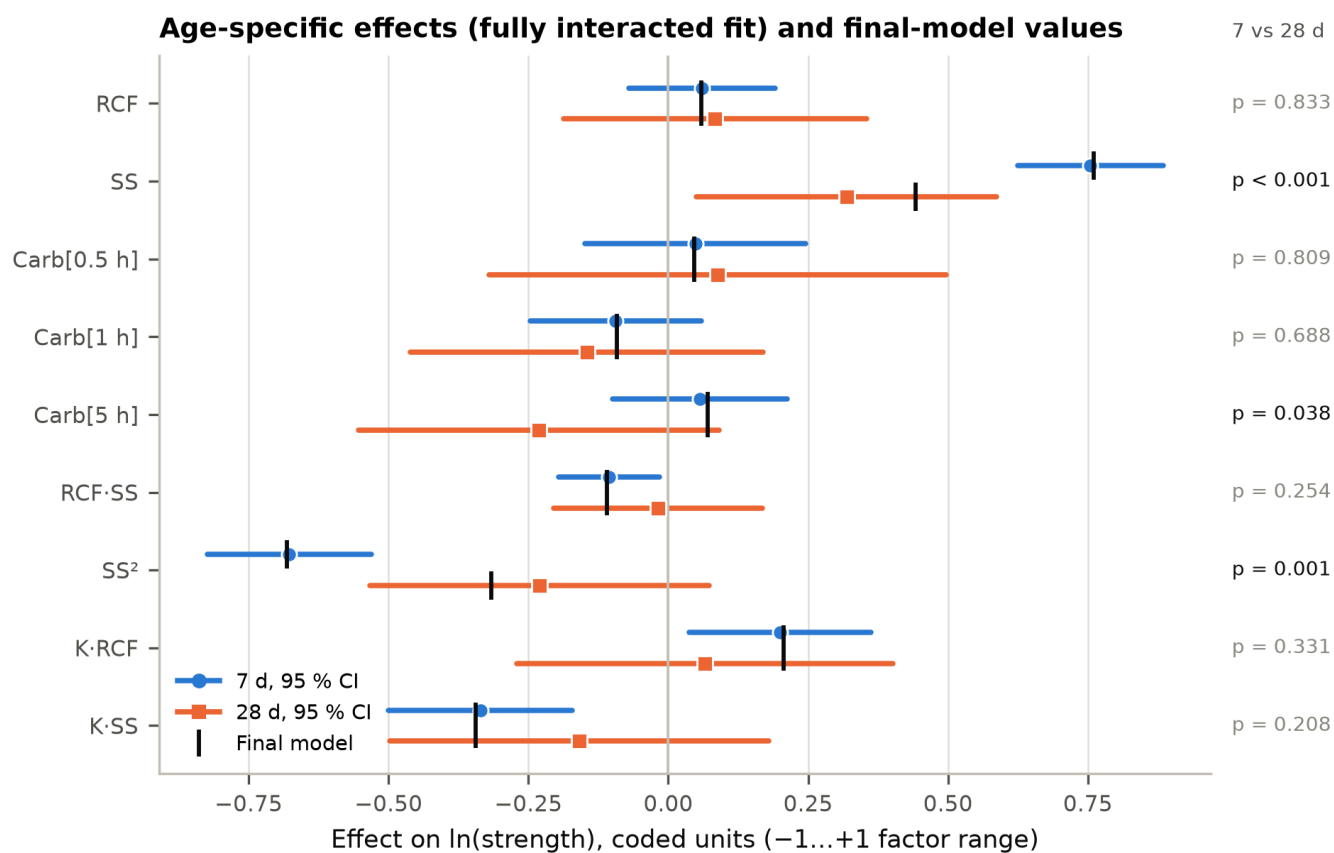
Wald F tests, Satterthwaite df. Covariance UN: SD 0.147 (7 d), 0.335 (28 d), $\rho = 0.52$. Lack of fit: gain $p = 0.24$, level|gain $p = 0.63$. Box-Cox (final): $\lambda = 0.14$ (95 % CI -0.10 to 0.41). Exhaustive search of 53,105 models; ΔAICc to best CS model 6.4.





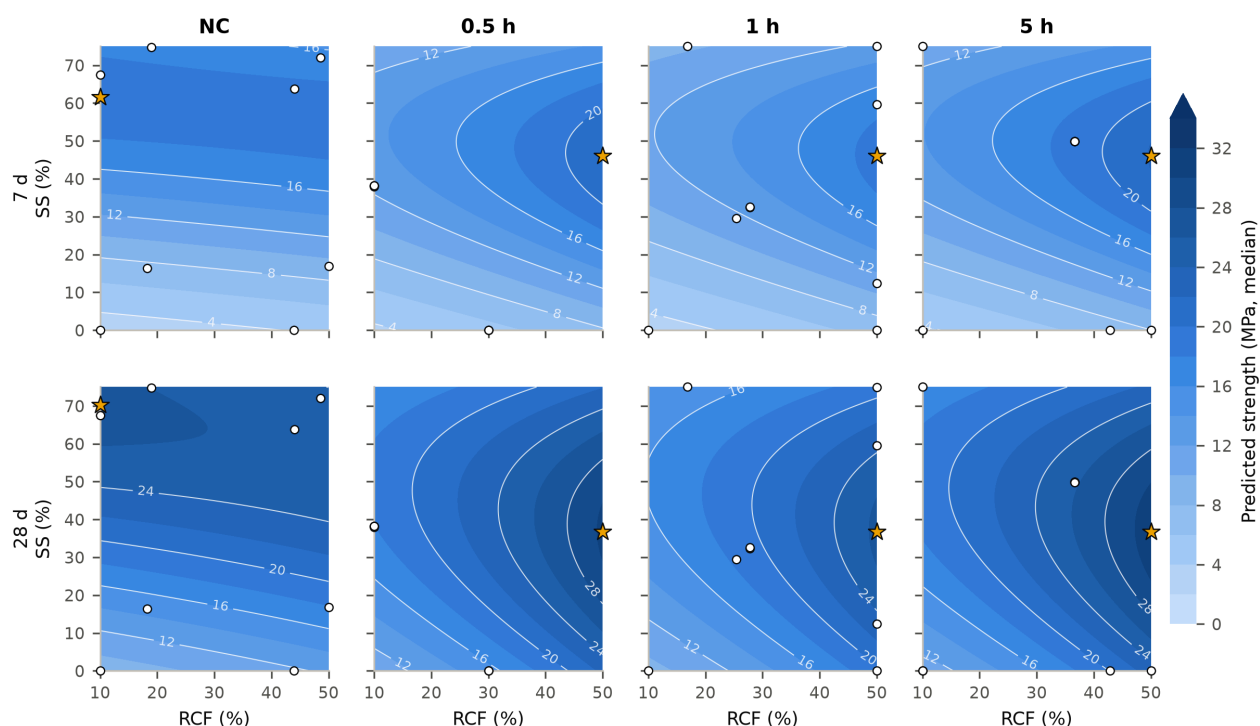
Age-specific effects

Term (coded)	Effect at 7 d [95 % CI]	Effect at 28 d [95 % CI]	Change 7→28 d	p (change)	Age term in final model
RCF	0.060 [-0.071, 0.191]	0.083 [-0.188, 0.355]	0.023 [-0.204, 0.250]	0.83	no (common effect)
SS	0.754 [0.625, 0.884]	0.319 [0.051, 0.587]	-0.436 [-0.660, -0.211]	< 0.001	yes
Carb 0.5 h	0.048 [-0.149, 0.246]	0.089 [-0.320, 0.497]	0.040 [-0.302, 0.382]	0.81	no (common effect)
Carb 1 h	-0.094 [-0.246, 0.059]	-0.145 [-0.460, 0.170]	-0.052 [-0.315, 0.212]	0.69	no (common effect)
Carb 5 h	0.057 [-0.099, 0.213]	-0.231 [-0.553, 0.091]	-0.288 [-0.557, -0.018]	0.038	no (common effect)
RCF·SS	-0.106 [-0.196, -0.016]	-0.018 [-0.204, 0.168]	0.088 [-0.068, 0.243]	0.25	no (common effect)
SS ²	-0.678 [-0.824, -0.531]	-0.230 [-0.533, 0.073]	0.448 [0.194, 0.701]	0.001	yes
K·RCF	0.199 [0.037, 0.362]	0.065 [-0.271, 0.401]	-0.134 [-0.415, 0.147]	0.33	no (common effect)
K·SS	-0.336 [-0.500, -0.172]	-0.159 [-0.498, 0.179]	0.177 [-0.107, 0.460]	0.21	no (common effect)



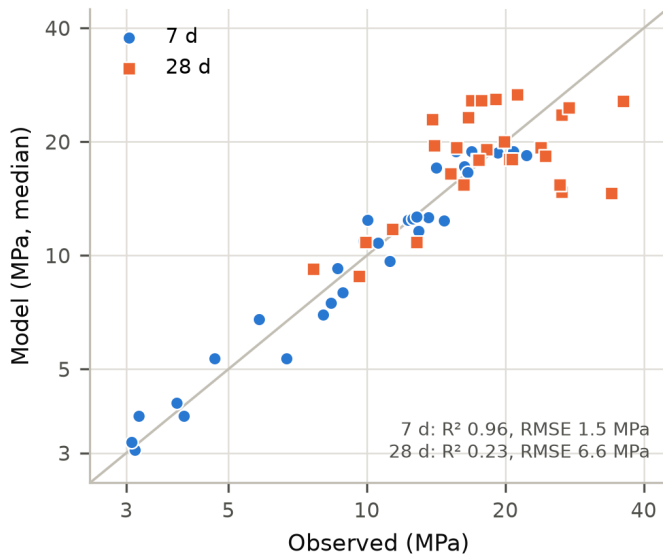
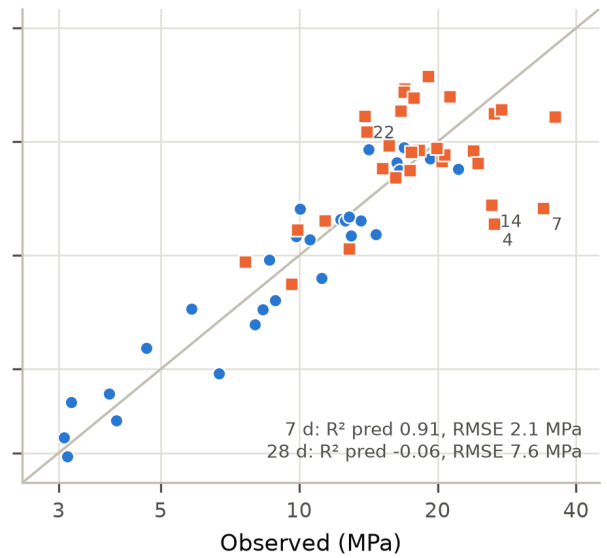
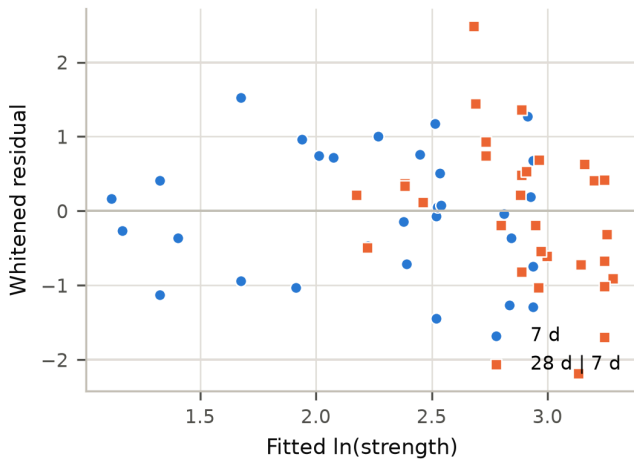
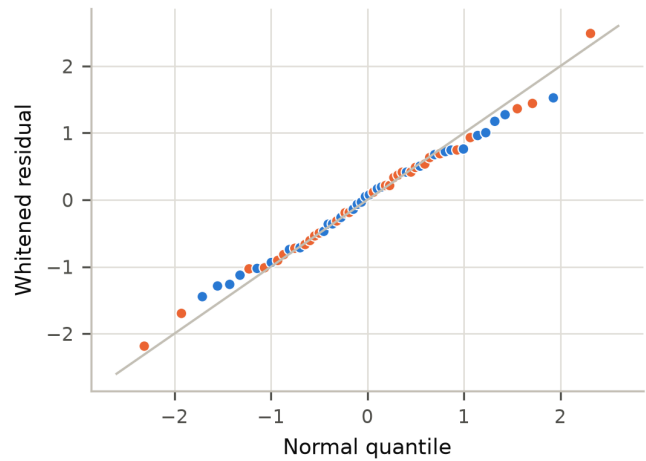
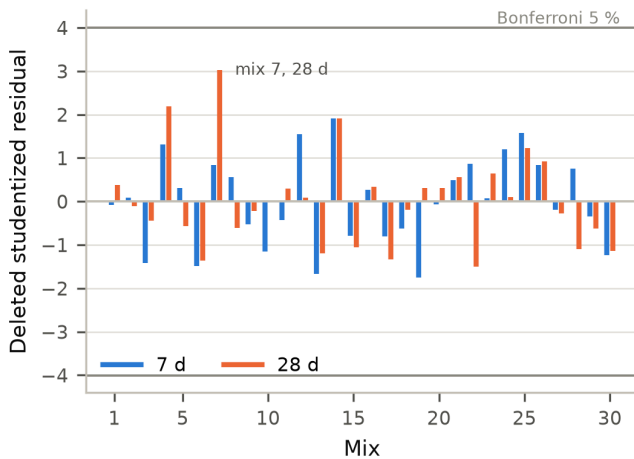
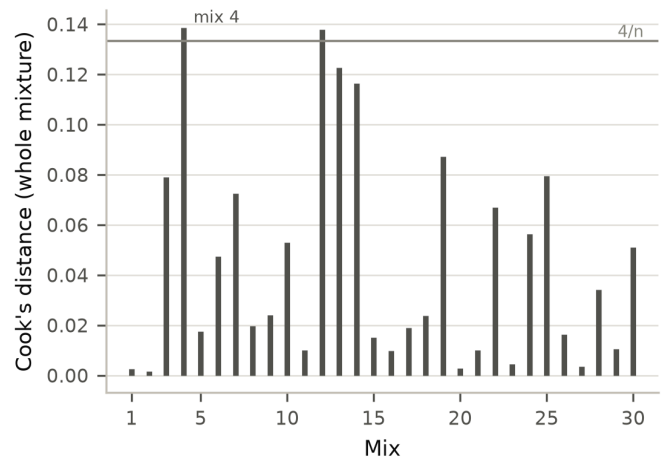
Age	Carbonation	Maximum at	Median (MPa)	95 % CI	95 % PI (one batch)	Within 5 % of maximum	Scaled pred. variance / design max
7 d	NC	RCF 10 %, SS 61.5 %	19.0	15.5–23.4	13.2–27.5	RCF 10–50, SS 46–72	0.45 / 0.59
7 d	0.5 h	RCF 50 %, SS 46.0 %	21.6	17.3–27.1	14.8–31.6	RCF 46–50, SS 36–56	0.54 / 0.59
7 d	1 h	RCF 50 %, SS 46.0 %	18.8	16.2–21.8	13.4–26.5	RCF 46–50, SS 36–56	0.23 / 0.59
7 d	5 h	RCF 50 %, SS 46.0 %	22.1	18.5–26.5	15.5–31.5	RCF 46–50, SS 36–56	0.34 / 0.59
28 d	NC	RCF 10 %, SS 70.2 %	26.7	19.9–35.8	12.7–56.1	RCF 10–50, SS 49–75	0.19 / 0.24
28 d	0.5 h	RCF 50 %, SS 36.8 %	30.4	22.8–40.4	14.5–63.6	RCF 46–50, SS 22–52	0.18 / 0.24
28 d	1 h	RCF 50 %, SS 36.8 %	26.5	20.8–33.8	12.8–54.8	RCF 46–50, SS 22–52	0.13 / 0.24
28 d	5 h	RCF 50 %, SS 36.8 %	31.1	24.0–40.3	15.0–64.6	RCF 46–50, SS 22–52	0.14 / 0.24

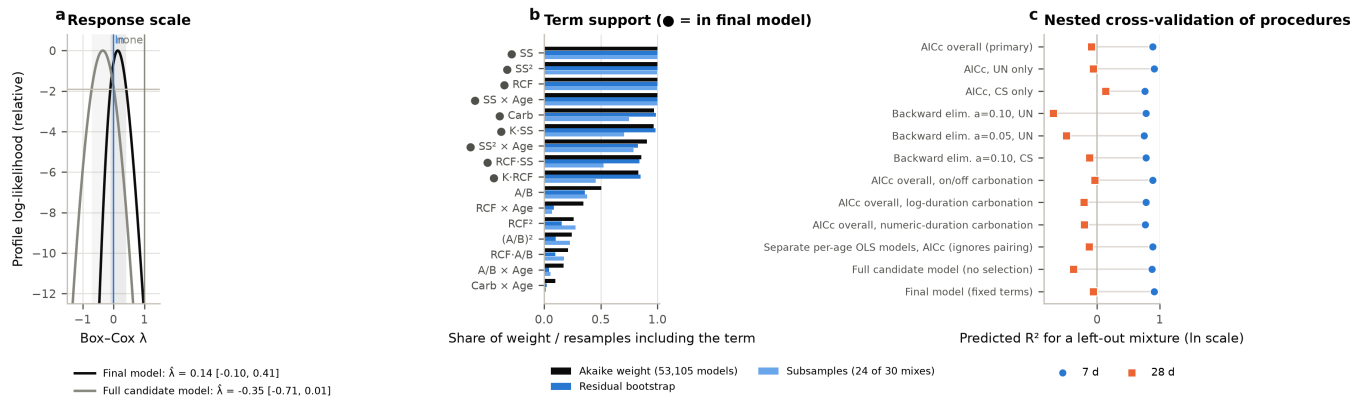
Predicted strength over RCF × SS (A/B not in model). ○ runs at that carbonation level, ★ maximum



Validation and robustness

Leave-one-mixture-out predicted R^2 : 0.91 (7 d), -0.06 (28 d); RMSE 2.1 / 7.6 MPa. Nested CV of the selection procedure: 0.89 / -0.09. Residual-bootstrap re-selection keeps SS as the only age-dependent variable in 86 % of resamples. Shapiro-Wilk $p = 0.99$; Breusch-Pagan $p = 0.62$; largest $|t|$ 3.02 (mix 7, 28 d; Bonferroni $p = 0.44$).

a Fitted (all 30 mixes)**b Leave-one-mixture-out prediction****a Residuals vs fitted (Breusch-Pagan $p = 0.62$)****b Normal Q-Q (Shapiro-Wilk $p = 0.99$)****c Outliers (mixture left out of the fit)****d Influence of each mixture on the coefficients**



Robustness: key model quantities under alternative analyses (black = primary)

